2.6.3 Additional Tests for Modified Asphalt Binders

In addition to conventional specification tests common for all asphalt binders, modified asphalt binders often have additional tests that are required to characterize their properties. These range from separation tests that characterize the compatibility of the asphalt-polymer blend to recovery and stress-strain tests that characterize the elastic behavior of the modified asphalt binder.

Separation Tests

Ointment Tube

A commonly used test to assess polymer-asphalt compatibility is generally referred to as the "ointment" or "cigar" tube test. In this test, an aluminum ointment tube of a specific size is filled with 50 grams of modified asphalt binder (see Figure 2.46). The filled tubes are then sealed and allowed to stand vertically in an oven operating at 325°F (163°C) for 48 hours. The tubes are then transferred to a freezer for four hours. After freezing, the tubes are removed and cut into thirds. The top third and bottom third are separated into tins and tested using either the softening point or DSR tests to determine if a significant difference exists in the properties of the top and bottom thirds. A significant difference in test values indicates that the polymer may have separated from the asphalt binder.

Solubility

Centrifuge

The solubility test described earlier was first introduced into a specification in 1903. The solubility test for petroleum asphalt binders, as it exists today in asphalt binder specifications, is used to ensure that an asphalt binder does not contain insoluble organic and inorganic matter that does not contribute to the "active cementing constituents" of the asphalt binder. As the use of polymer-modified asphalt binders has increased, a different solubility test (ASTM D 5546) was developed. ASTM D 5546 is similar to the standard solubility test except that it uses a centrifuge and toluene as the solvent. It was developed principally because some polymer modifiers did not dissolve adequately during the standard solubility test procedure, thus resulting in some polymer-modified asphalt binders failing the solubility criterion.

Recovery and Stress-Strain Tests

To evaluate whether an asphalt binder has been modified with an elastomer—specifically a block copolymer such as SBS—specifying agencies will often add tests designed to measure the ability of the asphalt binder to stretch when loaded and ultimately rebound to its original shape when unloaded. The ability, and extent, to which an asphalt binder will stretch and recover is related to the level of elastomeric modification in the asphalt binder.

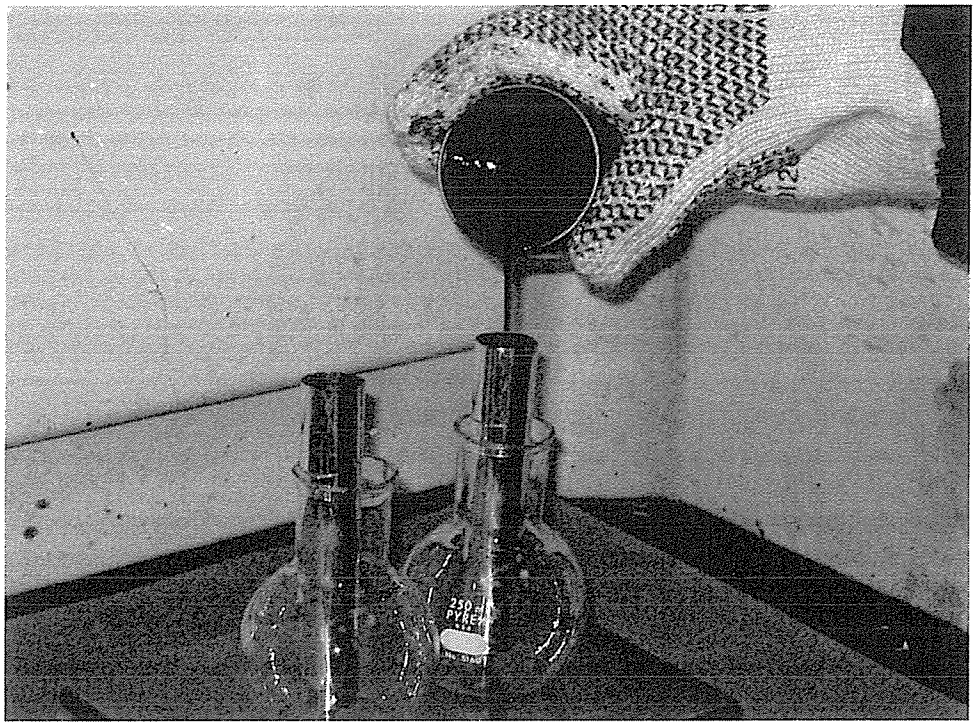


Figure 2.46 Preparing Separation Test Samples

Three tests are commonly used in the United States for testing elastomeric modified asphalt binders: elastic recovery, force ductility, and toughness and tenacity. Each test involves stretching an asphalt binder sample and measuring either the stress-strain response or recovery amount.

An additional parameter used by some states is the phase angle measured using the dynamic shear rheometer. The phase angle is an indicator of the relative amount of the elastic component in the asphalt binder.

An additional test proposed by the NCHRP 9–10 research for characterizing modified asphalt binders was the DSR Repeated Creep Test. This test has been modified as a multiple stress creep recovery (MSCR) test and is being extensively evaluated as a possible test that characterizes the behavior of modified asphalt binders.

Elastic Recovery

The Elastic Recovery test, as detailed in AASHTO T 301, is a simple procedure based on the ductility test equipment. In the test, a specimen is prepared as in the ductility procedure, except that the side pieces are flat, rather than V-shaped (see Figure 2.47).

To conduct the test, a heated sample of asphalt binder (usually RTFO-aged) is poured into the mold, slightly overfilling the mold. After a prescribed cooling period, the specimen is trimmed flush with the surface of the mold. The trimmed test specimen is placed in the ductility water bath and conditioned to the desired test temperature, usually 77°F (25°C). After the conditioning

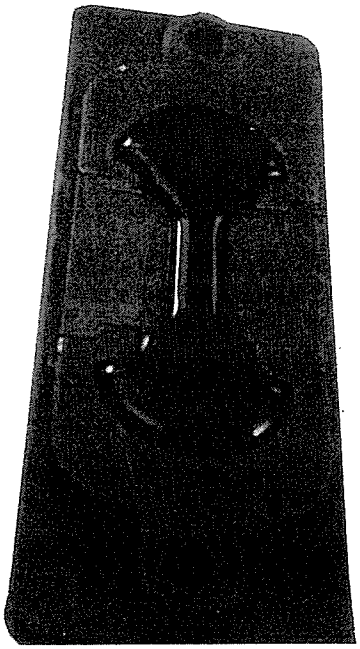


Figure 2.47 Elastic Recovery Test Specimen

period, the specimen is loaded into the ductility machine and one end of the specimen is pulled away from the other at a rate of 5 centimeters per minute until the specimen has stretched a prescribed length—either 10 or 20 centimeters (see Figure 2.48). Depending on the procedure, the specimen is then either immediately cut in the center or held in a stretched position for five minutes before cutting in the center. After one hour, the specimen is pushed back together until the cut ends touch. This final measurement is then used to calculate the recovery of the asphalt binder specimen. Higher recovery values are indicative of more elastic asphalt binders.

Force Ductility

Like the Elastic Recovery test, the Force Ductility test (as detailed in AASHTO T 300), is a simple procedure based on the ductility test equipment. In the test, a specimen is prepared as in the ductility procedure, except that the side pieces are flat, rather than V-shaped.

The specimen preparation and test procedure is essentially identical to the elastic recovery test. The main differences between the tests are the following:

- A load cell is used to record the force required to pull the asphalt binder specimen.
- The test is continued to 30 centimeters (see Figure 2.49) at which point the test ends.
- Test temperature is usually 41°F (4°C).
- The asphalt binder is usually tested in the original (unaged) condition.

In the test, a stress-strain curve is constructed (see Figure 2.50). Typically, the stress at 30 centimeters is divided by the peak stress to generate a

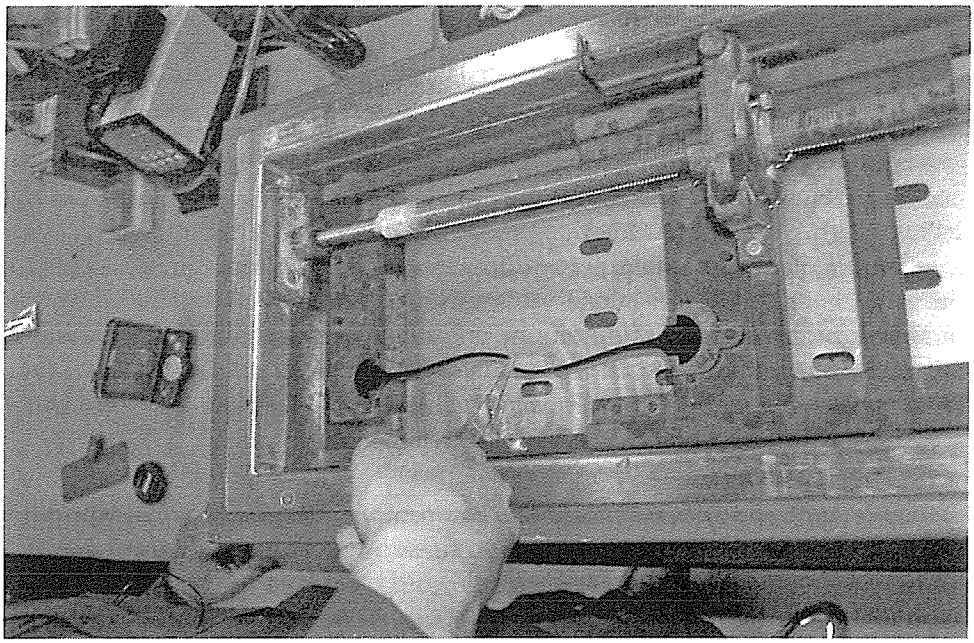


Figure 2.48 Elastic Recovery Test Operation

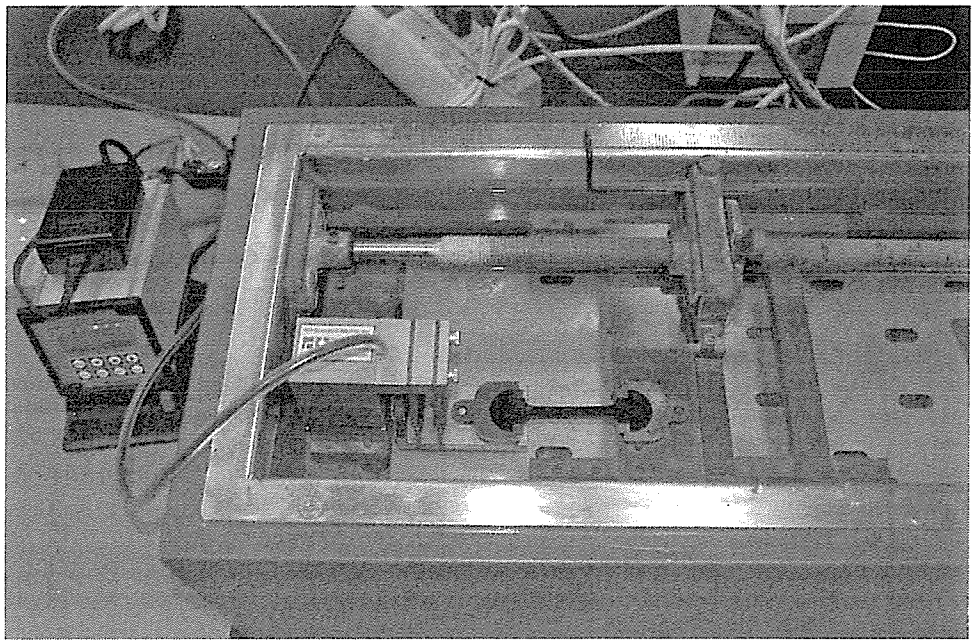


Figure 2.49 Force Ductility Test Operation

force ductility ratio. Specimens that do not stretch to 30 centimeters before breaking are considered to fail the test. Increasing force ductility ratios are indicative of increasing elasticity in the asphalt binder.

Toughness and Tenacity

A third test conducted by some agencies in the United States is the Toughness and Tenacity test. This procedure was first introduced by Benson in the 1950s. In the test, a metal hemispherical head is embedded in hot asphalt to a depth of approximately 11 millimeters. After cooling to 77°F (25°C), the head is attached to a tensile test machine and pulled from the asphalt binder at a rate of 51 centimeters per minute (see Figure 2.51). The load is measured throughout the test and a load-deformation curve is plotted (see Figure 2.52). Toughness and Tenacity values are determined based on the area under different portions of the load-deformation curve.

DSR Phase Angle

In the standard DSR test used for Performance Graded asphalt binders, the complex shear modulus (G^*) and phase angle (δ) are determined. The phase angle is a measure of the elastic and viscous components of the asphalt binder. Some user agencies specify a maximum phase angle for modified asphalt binders. Unfortunately, the phase angle is influenced by the stiffness of the asphalt binder. Thus, it is not a true reflection of the elasticity—in the form of an elastomer—of the asphalt binder.

DSR Creep-Recovery

A relatively new test being evaluated by researchers for characterizing the properties of modified asphalt binders is the DSR repeated creep-recovery test. This test, originally proposed by researchers in 2000, has been modified to include multiple stress levels. It is now referred to as the multiple-stress creep-recovery (MSCR) test.

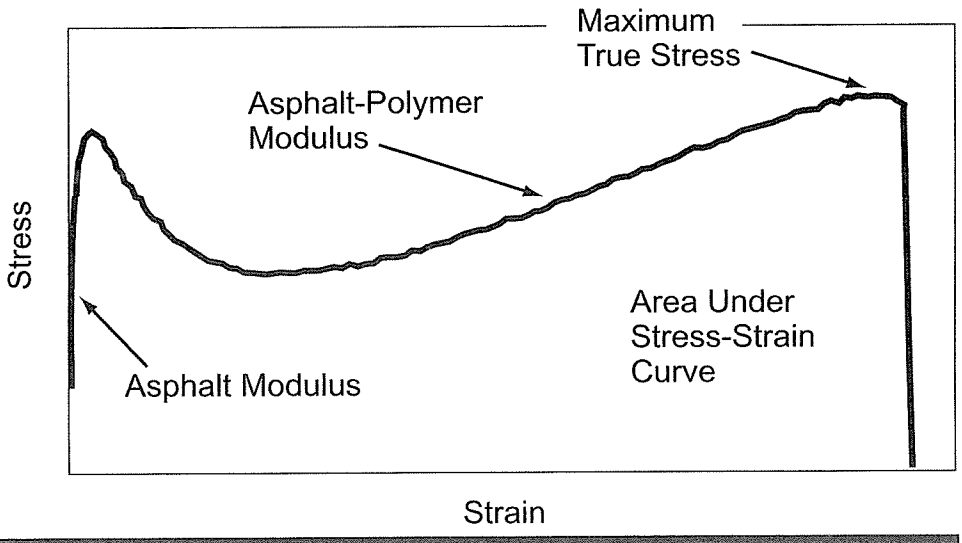


Figure 2.50 Force Ductility Output

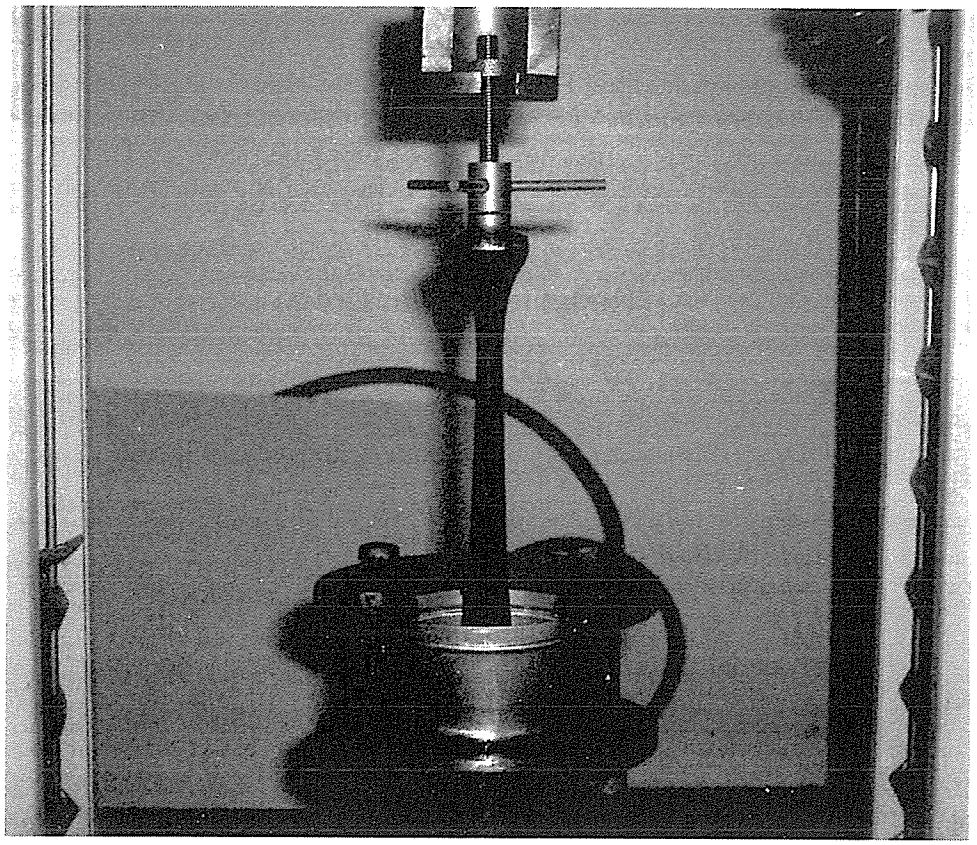


Figure 2.51 Toughness and Tenacity Equipment and Operation

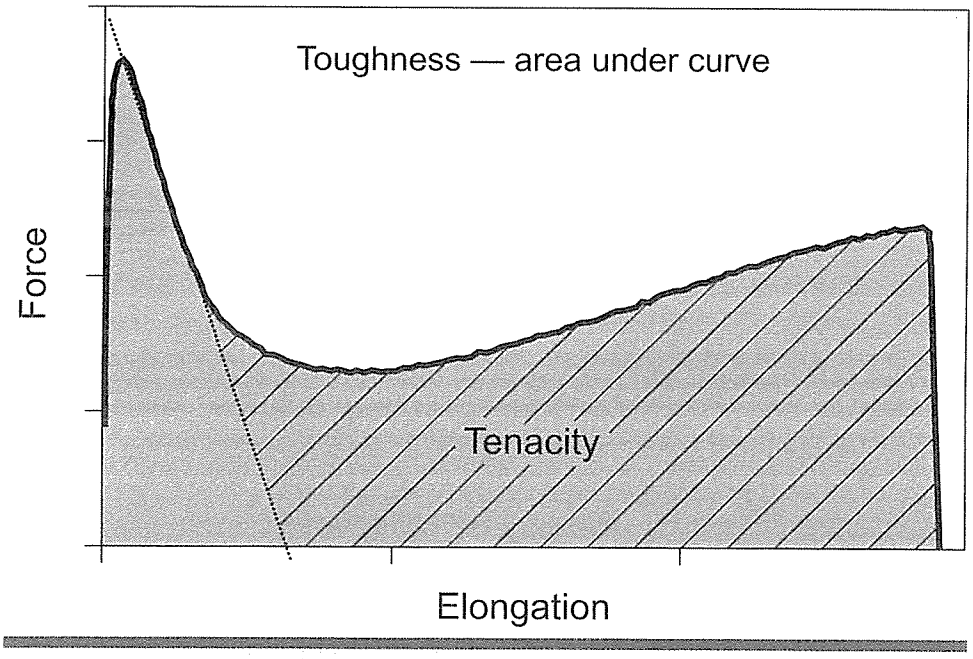


Figure 2.52 Toughness and Tenacity Output

In the test, a specimen is prepared and loaded in the DSR as if conducting measurements to obtain $G^*/\sin\delta$ values. The test then differs in that a specified shear stress is applied for a short time and then removed, allowing the specimen to recover from its deformed state (see Figure 2.53). The amount of recovery in each cycle is measured. The test may be conducted at different stress levels to evaluate the stress dependency of the modified asphalt binder.

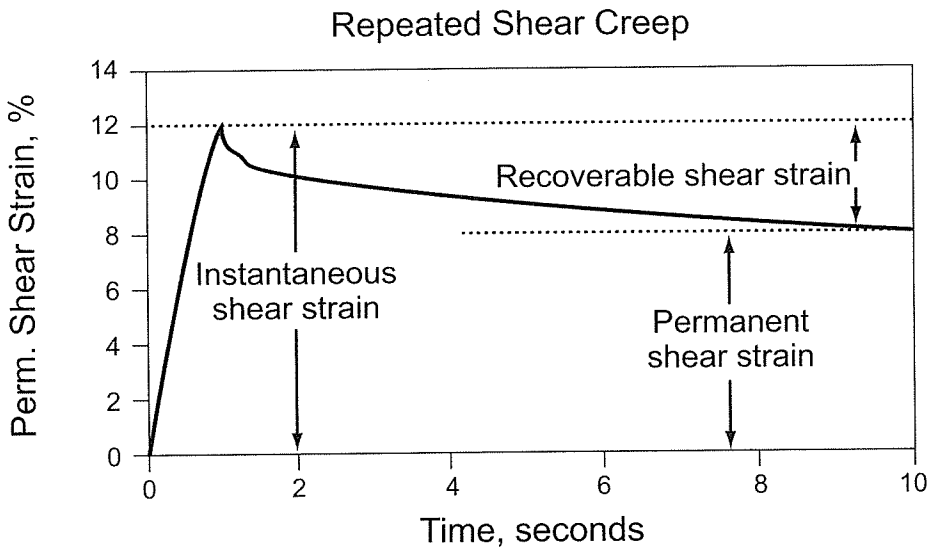


Figure 2.53 Creep Recovery Output

2.7 Supplementary Asphalt Topics

2.7.1 Control of Asphalt Temperatures

Asphalt is a thermoplastic material that decreases in viscosity with increasing temperature. The relationship between temperature and viscosity, however, may not be the same for different sources or types and grades of asphalt material.

Mixing Temperatures

Both asphalt and aggregate must be heated before they are mixed—the asphalt to make it fluid enough to be pumped and the aggregate to make it dry and hot enough to keep the asphalt fluid while it is coating the particles. The degree of heat in the dried aggregate controls the temperature of the asphalt-aggregate mixture because the temperature of the asphalt cement rapidly adjusts to that of the aggregate when the two are mixed.

The mixing temperature for a hot mix asphalt mixing facility is typically governed by placing and compaction requirements specific to the project. Best practice suggests that the lowest temperature that will allow time for the mixture to be hauled, placed, and sufficiently compacted should be used. Because asphalt binder properties and project conditions may vary widely, it is not appropriate to definitively recommend minimum or maximum mixing temperatures. However, to provide guidance, the Asphalt Pavement Environmental Council has published recommended ranges of storage and mixing temperatures at the hot mix asphalt mixing facility based on the grade of the asphalt binder (see Table 2.6). The user is encouraged to consider these recommended temperature ranges when conducting mixing and compaction operations. The user may also wish to discuss appropriate mixing temperatures with the asphalt binder supplier. Regardless, it is strongly recommended that the maximum mixing temperature should never exceed 350°F (177°C).

Temperatures for emulsified asphalt-aggregate mixtures and cutback asphalt-aggregate mixtures are substantially lower than those required for hot mix asphalt. While these lower mixing temperatures may not provide thoroughly dry aggregates, experience has shown that they can be satisfactory for the intended purpose, especially when asphalt emulsions are used. Temperatures of 50–158°F (10–70°C) are typically used for mixing with emulsified asphalts. Typical temperatures for mixing with MC and SC cutback asphalts may range from 131°F to 239°F (55°C to 115°C) depending on the grade. RC cutback asphalts are not recommended for use in hot mixing. In any case, caution should be used when selecting mixing temperatures for cutback asphalt-aggregate mixtures in the event that the selected temperature is above the flash point of the material.

Table 2.6
Ranges of Storage and Mixing Temperatures

Asphalt Pavement Environmental Council Best Practices

Typical Asphalt Binder Temperatures

Binder Grade	HMA Plant Asphalt Tank Storage Temperature° (F)		HMA Plant Mixing Temperature° (F)	
	Range	Midpoint	Range	Midpoint
PG 46–28	260–290	275	240–295	264
PG 46–34	260–290	275	240–295	264
PG 46–40	260–290	275	240–295	264
PG 52–28	260–295	278	240–300	270
PG 52–34	260–295	278	240–300	270
PG 52–40	260–295	278	240–300	270
PG 52–46	260–295	278	240–300	270
PG 58–22	280–305	292	260–310	285
PG 58–28	280–305	292	260–310	285
PG 58–34	280–305	292	260–310	285
PG 64–22	285–315	300	265–320	292
PG 64–28	285–315	300	265–320	292
PG 64–34	285–315	300	265–320	292
PG 67–22	295–320	308	275–325	300
PG 70–22	300–325	312	280–330	305
PG 70–28	295–320	308	275–325	300
PG 76–22	315–330	322	285–335	310
PG 76–28	310–325	318	280–330	305
PG 82–22	315–335	325	290–340	315

Use mid-point temperature for test strip construction.

Spraying Temperatures

The lower temperatures are more critical for spraying asphalt cements and cutback asphalts; therefore, only minimum temperatures are recommended for these asphalt materials. The highest temperature used for spraying asphalt cement should be that at which no fogging occurs when the asphalt material leaves the spray nozzle.

For asphalt cements, typical minimum spraying temperatures range from 266°F to 302°F (130°C to 150°C). For emulsified asphalts, typical minimum spraying temperatures range from 68°F to 185°F (20°C to 85°C). For cutback asphalts, typical minimum spraying temperatures range from 68°F to 230°F (20°C to 110°C). As noted previously, the selected application temperature may, in some cases, be above the flash point of the material. Caution must therefore be exercised to prevent fire or an explosion.

Precautions

The temperature ranges discussed in the preceding paragraphs are necessary to provide proper asphalt viscosity for spraying and mixing

applications for the various grades of asphalt cement, emulsified asphalt, and cutback asphalt. It must be recognized, however, that some of the temperature ranges indicated may be above the minimum flash point for the RC, MC, and SC cutback asphalt materials. In fact, some asphalts will "flash" at temperatures below these indicated ranges. Accordingly, suitable safety precautions are mandatory at all times when handling cutback asphalts. These safety precautions include, but are not limited to, the following:

- Open flames or sparks must not be permitted close to these materials. Controlled heat should be applied in heating kettles, mixers, distributors, or other equipment designed and approved for the purpose.
- Open flames must not be used to inspect or examine drums, tank cars, or other containers in which these materials have been stored.
- All vehicles transporting these materials must be properly vented.
- Only experienced personnel must be permitted to supervise the handling of these materials.
- All applicable intrastate and interstate commerce requirements must be met.

2.7.2 Asphalt Sampling

To be meaningful, a sample of shipped asphalt must be truly representative, and it must be handled carefully so that it does not become contaminated or altered in any way before being tested. If the sample is not truly representative, test results can be misleading.

Sampling Asphalt Binder from Vehicle Tanks

The best way to obtain a representative sample of asphalt from a vehicle tank is to use a sampling valve in the truck tank. If the sampling valve is used properly, representative samples can be obtained easily and quickly.

The procedure is a simple matter of opening the valve, allowing at least 1 quart (1 liter) of material to be drained off to remove prior material, and then completely filling the sample container. Certain precautions must be observed, however, to prevent contamination. Sample containers must be new, clean, and dry. The filled sample container must be tightly and positively sealed immediately after the sample is obtained. The sample container should not be submerged in solvent or wiped with a solvent-saturated cloth. Any spilled material on the outside of the container should be wiped with a clean, dry cloth immediately after the container has been sealed.

Because the asphalt in most cases is hot, safety precautions are mandatory at all times during sampling. Gloves should be worn, with sleeves rolled down and fastened over the gloves while sampling and sealing containers. Face shields are also a must, and no smoking should be permitted. Tongs or some such device should be used to hold the sample container while the sample is being obtained, and the sampler should stand above

and away from the sampling valve to the windward side. Take the sample slowly and carefully to prevent the hot material from splashing.

Immediately after filling, sealing, and cleaning as required, the sample containers are marked for identification.

Contamination

It is extremely important that contamination of asphalt products be prevented—from the time they leave the refinery to their end use. This importance also extends to samples and sampling procedures. Tests carried out to ensure compliance with specifications are only meaningful when they are performed on samples that truly represent the material used.

Contributing causes for asphalt binders failing to meet specification requirements can be quite numerous, but contamination is the most common culprit. Investigations have shown that many off-specification samples have flash points that are too low, penetration values that are too high, viscosities or stiffness that are too low, or erratic distillation test results. All of these responses are typical of solvent-type contaminations. For example, experiments have indicated that as little as 0.1 percent of diesel in an asphalt cement (equivalent to approximately 1 gram in a full quart can) may lower the flash point by as much as 50°F (28°C), and increase the penetration as much as 10 points. Such contamination appears to be minimal, but the resulting effect on the physical properties of the asphalt binder may be substantial.

Vehicle Tank Conditions for Loading Asphalt

A log showing the material hauled on the previous load must be maintained by the hauler. The log should show whether the tank was cleaned or only drained after the last load. If requested, the hauler should certify that the tank is suitable to load with the product specified. The pump, unloading line, and all piping on the truck must be emptied. Adequate provision must be made for taking a representative sample of the tank contents after loading is completed. A dip sample from the surface of the freshly loaded product is not normally representative.

2.7.3 Temperature-Volume Relationship

Density and Specific Gravity

The *density* of a material is its mass per unit volume measured at a specific temperature. It is usually given in terms of *grams per cubic centimeter, kilograms per cubic meter, or pounds per cubic foot.* The *specific gravity* of a material is the ratio of the density of a material to the density of water, both measured at a specified temperature. It is a value that has no units. Because the density of water is 1 gram per cubic centimeter at 77°F (25°C), it is convenient to measure the density of a material at 77°F (25°C) in grams per cubic centimeter. By doing this, the specific gravity of the material is determined by dividing the density by 1.

Asphalt binder has a density, or specific gravity, very similar to water. Like water, the specific gravity of an asphalt binder changes with temperature. Additionally, different grades of asphalt binders have different specific gravity values. Softer grades usually have a lower specific gravity than stiffer (more viscous) grades.

There is no specification requirement for the density or specific gravity of an asphalt binder because, like temperature-viscosity relations, it varies with different asphalt binders. However, there often is need to know the exact density, or specific gravity, of an asphalt binder. This value is used, for example, in computing the volume of asphalt at certain temperatures for purposes of purchase payments, or in designing asphalt-aggregate mixtures.

For paving-grade asphalt binders and viscous cutback asphalts, the test is usually performed with a pycnometer bottle as previously described. For more fluid asphalt binders and asphalt emulsions, a hydrometer is used.

To indicate definite conditions applicable to a given specific gravity value, the temperature of the asphalt and water should be shown. Thus a specific gravity at 60/60°F (15.6/15.6°C) indicates that the determination was made with both materials at a temperature of 60°F (15.6°C). The specific gravity of asphalt at 60/60°F (15.6/15.6°C) can be calculated knowing the specific gravity at 77/77°F (25/25°C). This procedure is explained in the following paragraphs.

Temperature-Volume Relationship and Calculations

All liquids and most solids undergo changes in volume with changes in temperature. They expand when heated and contract when cooled. The change in unit volume per change in unit temperature is called coefficient of expansion, a factor that varies with variations in the density (specific gravity) of the asphalt product.

Temperature-volume correction tables for asphalt materials are published in ASTM D 4311. The multipliers given in the tables are used to convert a known volume at a given temperature to volume at 60°F (15.6°C), which is customarily used as the standard basis for volume determinations of asphalt. The equation is as follows:

$$V = V_T \times M_T \quad \text{(Equation 1)}$$

Where

V = volume at 15.6°C or 60° F
V_T = volume at given temperature
M_T = multiplier from appropriate table

Example (U.S. Customary Units)

The specific gravity of an asphalt product is determined to be 0.985 at 60°F. The volume of this material is measured to be 9000 gallons at 180°F. Using the data in ASTM D 4311, the correction factor, M_T, for a temperature of 180°F is 0.9587. Thus, the volume of the material at 60°F is 9000 × (0.9587) = 8,628 gallons.

Specific Gravity Calculations

The basic formula for specific gravity of a material is the mass in air of a unit volume of the material at a specific temperature divided by the mass in air of an equal volume of water at the same temperature. This is written as follows:

$$G_x = \frac{W_x}{W_w} \quad \text{(Equation 2a) or } G_x = \frac{W_x}{V_x \times \gamma_w} \quad \text{(Equation 2b)}$$

where

G_x = specific gravity of the substance
W_x = mass of a unit volume of the substance
W_w = mass of a unit volume of water
V_x = volume of the substance
γ_w = density of water

The volume of asphalt, which changes with temperature, can be determined using the equation. However, the density of water also changes with temperature.

If the specific gravity of asphalt is found at 25/25°C (77/77°F), its specific gravity at 15.6/15.6°C (60/60°F) can be calculated by expressing *Equation 2b* for both temperatures, combining, appropriately substituting *Equation 1* and simplifying to become:

$$G_{15.6/15.6} = \frac{G_{25/25} \times \gamma_{w25}}{M_{25} \times \gamma_{w15.6}} \text{ or } G_{60/60} = \frac{G_{77/77} \times \gamma_{w77}}{M_{77} \times \gamma_{w60}} \quad \text{(Equation 3)}$$

where

$\gamma_{w25}(\gamma_{w77})$ = density of water at 25°C (77°F) = 0.9970 g/ml
$\gamma_{w15.6}(\gamma_{w60})$ = density of water at 15.6°C (60°F) = 0.9988 g/ml
M_{25} (M_{77}) = multiplier at 25°C (77°F) from ASTM D 4311

As an example, assume the specific gravity of an asphalt cement at 77/77°F is 1.003. From ASTM D 4311, the correction factor, M_T, for 77°F is 0.9941. The specific gravity at 60/60°F is then

$$G_{60/60} = \frac{G_{77/77} \times \gamma_{w77}}{M_{77} \times \gamma_{w60}} = \frac{1.003 \times 0.9970}{0.9941 \times 0.9988} = 1.007$$

Tank Measurements

Many asphalt containers for transporting and storage are cylindrical tanks that are in a horizontal position. Determination of the volume of material in the tank consists of measuring its depth. Table 2.7 gives quantities in terms of percent of the total capacity based on the depth in percent of the diameter.

Volumes of asphalt are usually expressed in gallons (liters), and the formula for determining the capacity of a cylindrical tank is shown in Equation 4a for metric units (liters, meters) and Equation 4b for U.S. Customary Units (gallons, feet):

$$V = 785 \times D^2 \times L \quad \text{(Equation 4a) or } V = 5.88 \times D^2 \times L \quad \text{(Equation 4b)}$$

where

V = volume, liters or gallons
D = diameter of interior of tank, meters or feet
L = length of interior of tank, meters or feet

Example: (U. S. Customary Units)

Assume a horizontal tank has a capacity of 12,740 gallons and a diameter of 9 feet 6 inches (9.5 feet). If the depth of the asphalt in the tank is 7 feet 3.5 inches (7.29 feet), then the percent depth filled is

$$\text{Percent Depth Filled} = \frac{\text{Depth}}{\text{Diameter}} \times 100 = \frac{7.29}{9.5} \times 100 = 76.8 \text{ percent}$$

By interpolating in Table 2.7, the volume of asphalt in the tank is

$$\text{Volume} = \frac{\text{Tank Capacity} \times \% \text{ Capacity}}{100} = \frac{12{,}740 \times 82.40}{100} = 10{,}498 \text{ gallons}$$

Table 2.7
Percent Capacities for Various Depths of Cylindrical Tanks in Horizontal Position

Percent Depth Filled	Percent of Capacity	Percent Depth Filled	Percent of Capacity	Percent Depth Filled	Percent of Capacity	Percent Depth Filled	Percent of Capacity
1	0.20	26	20.73	51	51.27	76	81.50
2	0.50	27	21.86	52	52.55	77	82.60
3	0.90	28	23.00	53	53.81	78	83.68
4	1.34	29	24.07	54	55.08	79	84.74
5	1.87	30	25.31	55	56.34	80	85.77
6	2.45	31	26.48	56	57.60	81	86.77
7	3.07	32	27.66	57	58.86	82	87.76
8	3.74	33	28.84	58	60.11	83	88.73
9	4.45	34	30.03	59	61.36	84	89.68
10	5.20	35	31.19	60	62.61	85	90.60
11	5.98	36	32.44	61	63.86	86	91.50
12	6.80	37	33.66	62	65.10	87	92.36
13	7.64	38	34.90	63	66.34	88	93.20
14	8.50	39	36.14	64	67.56	89	94.02
15	9.40	40	37.39	65	68.81	90	94.80
16	10.32	41	38.64	66	69.97	91	95.55
17	11.27	42	39.89	67	71.16	92	96.26
18	12.24	43	41.14	68	72.34	93	96.93
19	13.23	44	42.40	69	73.52	94	97.55
20	14.23	45	43.66	70	74.69	95	98.13
21	15.26	46	44.92	71	75.93	96	98.66
22	16.32	47	46.19	72	77.00	97	99.10
23	17.40	48	47.45	73	78.14	98	99.50
24	18.50	49	48.73	74	79.27	99	99.80
25	19.61	50	50.00	75	80.39		

2.7.4 Frequently Asked Asphalt Questions

Some material properties of asphalt binders, other than those previously described, are often the subject of questions (for various reasons). A sampling of these questions are addressed in this section.

What Is the Thermal Conductivity of Asphalt Binder?

To calculate the thermal conductivity of asphalt binder, the Asphalt Institute has used the following equation:

$$K = (0.813/d)\times(1-(0.0003\times(t-32)))$$

where:

d = the specific gravity at 60°F/60°F
t = the temperature in F
K = the thermal conductivity (BTU-in)/(hour-foot^2-F)

This equation can be found in *Asphalts and Allied Substances: Volume IV,* by Abraham, which also has references for the equation.

What Is the Vapor Pressure of Asphalt at Typical Storage Temperatures?

It is estimated that at a typical inventory temperature of 325°F (163°C), the vapor pressure of petroleum asphalt is less than 0.01 pounds per square inch (1.5E–03 kiloPascals).

What Is the Typical Thermal BTU Value for a Pound of Asphalt?

The BTU value varies by temperature and by the percentage of mineral matter in the asphalt. Therefore, a range is usually quoted, but an approximate value of 158,500 BTU/gallon may be considered an average value typical of an AC-10 (or PG 58) grade. Most refineries will have a calorimeter in the lab that may be used to determine this value. Actual test results should be obtained if a more accurate value is needed.

What Is the Typical Value for the Specific Heat of Asphalt?

A typical value for the specific heat of asphalt is 0.22 calorie/gram-°C.

2.8 Safety Procedures for Hot Asphalt Binder

Because asphalt products are often stored and handled at elevated temperatures, fire prevention is extremely important. One of the greatest hazards in handling hot asphalt is exposure to a source of ignition. Sources of ignition, such as sparks of electrical or other origin, open flames or incandescent material (lighted cigarette) should be prohibited or otherwise strictly controlled in the vicinity of asphalt operations.

Many of the metal surfaces in asphalt plants and in roofing or processing operations exceed 150°F (65°C), and frequently are well over 200°F (93°C). Asphalt temperatures commonly exceed 300°F (149°C) as used in these areas. Exposed lines or surfaces can burn flesh on momentary contact at 150°F (65°C) or higher. Also, similar hazards exist in other areas: around dryers, boiler houses, asphalt receiving and handling areas, and similar locations. It is very important that workers be aware of burn hazards in the workplace. Burn hazards should be clearly indicated by warning signs. Workers should always wear appropriate protective clothing and follow proper safety procedures. Use catwalks, screens, barrier guards, and shields to protect from steam, hot asphalt, hot surfaces, and similar dangers.

Never enter an empty asphalt tank unless it is properly ventilated and has been purged of asphalt fumes. It is advisable to have a second person in the immediate proximity as an observer to be assured that one does not lose consciousness due to lack of oxygen.

Whenever a person is injured from exposure to asphalt fumes or hot asphalt, obtain first aid/medical attention immediately. If hot asphalt comes in contact with the skin, the affected area should be immersed in cool water. Do not attempt to wipe off the hot asphalt with a cloth because this may remove skin along with the asphalt. To prevent the possibility of future medical complications, have the victim examined by a physician even if the injury does not appear to be serious.

From a safety standpoint, it is desirable to store asphalt at a temperature as far as possible below the flash point. However, it should be recognized that the values reported by the flash point tests are specific to the test procedures employed and not necessarily representative of the vapor space atmospheres existent in storage.

Storage temperatures in the 212–266°F (100–130°C) range are particularly troublesome. Storage temperature should either be below 212°F (100°C) or above 266°F (130°C). In particular, temperatures should not be allowed to cycle above and below the boiling point of water (approximately 212°F or 100°C, depending on elevation).

For more information, please see the Asphalt Institute's VA-26, *Safe Handling of Hot Asphalt.*

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CHAPTER 3

Mineral Aggregates and Mixture Additives

This chapter provides information on aggregate as a component of asphalt mixtures and treatments. The sources of aggregates and production and processing practices are described. Aggregate properties that influence asphalt performance are covered. The basics of using reclaimed asphalt pavement (RAP) are summarized. Finally, information pertaining to aggregate additives is provided.

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- 3.1.1 Introduction
- 3.1.2 Origin of Rocks
- 3.1.3 Sources of Aggregates
- 3.1.4 Aggregate Production
- 3.1.5 Aggregate Properties

Section 3.2 Composite Aggregate Analysis
- 3.2.1 Introduction
- 3.2.2 Gradation

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- 3.3.1 Introduction
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- 3.4.1 Introduction
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- 3.4.3 Antistrip Additives
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3.1 Aggregates for Hot Mix Asphalt (HMA)

3.1.1 Introduction

The amount of aggregate in asphalt paving mixtures is generally 90–95 percent by weight or 75–85 percent by volume. Aggregate is primarily responsible for the load-supporting capacity of pavement; accordingly, performance of a hot mix asphalt (HMA) mixture is heavily influenced by the aggregate.

The general term *aggregate* refers to any hard, inert mineral material used for mixing in graduated particles or fragments. ASTM D 8 defines *aggregate* as "a granular material of mineral composition such as sand, gravel, shell, slag, or crushed stone, used with a cementing medium to form mortars or concrete or alone as in base courses, railroad ballasts, etc."

Aggregates can be either natural or manufactured. Natural aggregates are generally fragments extracted and processed from rock formations or rock deposits. Manufactured aggregates are derived from raw materials that have been physically or chemically changed or from industrial by-products that have been processed.

3.1.2 Origin of Rocks

The purpose of this section is not to describe in detail the origin of bedrock, from which most aggregate material is derived. A brief discussion is in order, however, because knowledge of the origin of rock is important for understanding the production and performance of mineral aggregates.

Rocks are divided into three general classifications: sedimentary, igneous, and metamorphic. These classifications describe how rocks are formed. Table 3.1 is a breakout of aggregate types and families for these three classifications. Table 3.2 shows some typical properties of four of the more common aggregate families.

Sedimentary Rocks

Sedimentary rocks are formed by the accumulation of sediment deposited by water or wind. With sufficient overburden, the sedimentation re-solidifies more or less to solid rock. The sediment may consist of rock or mineral fragments or particles of various sizes (conglomerate, sandstone, shale), the remains or products of animals or plants (certain limestone and coal), the product of chemical action or evaporation (salt, gypsum), or mixtures of these materials. Some sedimentary deposits consist of particles blown from volcanoes and deposited on land or in water. A characteristic feature of sedimentary deposits is a layered structure. This stratification is a direct result of the method of formation—the material having been deposited over some lake or sea bottom.

Table 3.1
Different Aggregate Classifications

Class	Type	Family
Sedimentary	Calcareous	Limestone Dolomite
	Siliceous	Shale Sandstone Chert Conglomerate[1] Breccia[1]
Igneous	Intrusive (coarse-grained)	Granite[2] Syenite[2] Diorite[2] Gabbro Periodotite Pyroxenite Hornblendite
	Extrusive (fine-grained)	Obsidian Pumice Tuff Rhyolite[2,3] Trachyte[2,3] Andesite[2,3] Basalt[2] Diabase
Metamorphic	Foliated	Gneiss Schist Amphibolite Slate
	Nonfoliated	Quartzite Marble Serpentinite

[1] May also be composed partially or entirely of calcareous materials.
[2] Frequently occurs as a porphyritic rock.
[3] Included in general term *felsite* when constituent minerals cannot be determined quantitatively.

Table 3.2
Typical Properties of Common Aggregates

Property	Granite	Limestone	Quartzite	Sandstone
Unit Weight (pcf)	162–172	117–175	165–170	119–168
Compressive Strength ($\times 10^3$ psi)	5–67	2.6–28	16–45	5–20
Shear Strength ($\times 10^3$ psi)	3.7–4.8	0.8–3.6	N/A(1)	0.3–3.0
Water Absorption (% by wt)	0.07–0.30	0.50–24.0	0.10–2.0	2.0–12.0
Specific Gravity	2.60–2.76	1.88–2.81	2.65–2.73	2.44–2.61

Note: (1) N/A = Data not available

Igneous Rock

Igneous rocks are formed by the cooling and solidification of hot molten materials (magma) that erupt from or are trapped beneath the earth's crust. These rocks are further classified as extrusive or intrusive rocks. Extrusive rocks are formed by material that has poured out on the earth's surface, such as that produced by volcanic action, and are distinguished in general by their glass-like structure (rhyolite, andesite, basalt, etc.). Intrusive rocks result from cooling and solidification at great depths within the earth's crust. They are wholly crystalline in texture (granite, diorite, gabbro, etc.). Although originally formed beneath the earth's surface, they are often exposed today as a result of earth movements and erosion processes.

Metamorphic Rocks

Metamorphic rocks are generally formed by the modification of sedimentary or igneous rocks because of intense pressures caused by movement within the earth's crust and by excessive heat and solutions. Factors causing such modifications are complex, and the original form of the altered rock is often difficult to determine. One distinguishing feature of metamorphic rocks is the parallel planes in which the minerals that form the rocks are arranged. This arrangement is called foliation. These planes are zones of weakness along which the rocks may be split more easily than in other directions. Examples of foliated rocks are gneisses and schists (altered from igneous material), and slate (altered from shale). Marble (altered from limestone) and quartzite (altered from sandstone) are common types of metamorphic rocks having a massive texture, usually without foliation.

3.1.3 Sources of Aggregates

Aggregates for asphalt paving are generally classified according to their source and means of preparation. Figure 3.1 shows the sources of aggregates throughout the United States. They include pit or bank-run aggregates, quarried or crushed stone, and synthetic and lightweight aggregates.

Pit or Bank-Run Materials

Gravel and sand are natural aggregates existing in loose or unconsolidated alluvial deposits and are typically pit or bank-run material extracted without drilling and blasting. *Gravel* is usually defined as aggregate particles ranging in size from 3 inches (75 millimeters) to No. 4 (4.75 millimeters). Sand consists of particles ranging in size from No. 4 (4.75 millimeters) to No. 200 (75 microns). Sizes No. 200 and smaller include the silt fraction. Many pit or bank-run alluvial or glacial sources also contain cobbles with aggregate particles as large as 10 inches (254 millimeters) or more.

Exposed rocks are eroded and degraded by many processes of nature, both physical and chemical. The products of the degradation processes are usually moved by wind, water, or moving ice, and deposited as a soil mineral in various landforms.

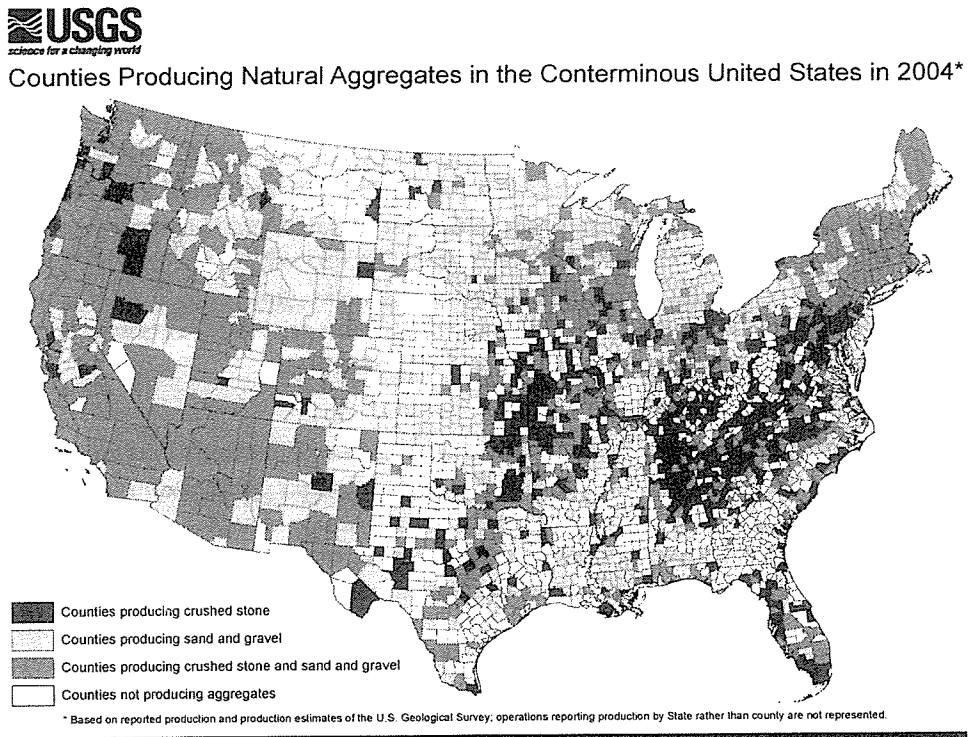


Figure 3.1 Sources of Natural Aggregates in U.S.

Gravels are widely distributed, but deposits are rarely found without some proportion of sand and possibly silt. Sandy soil mixtures frequently contain some clay and silt. Beach sands, some of which are now far inland, are a uniform-size material, but river sands often contain large amounts of gravel, silt, and clay.

Transporting agents also influence the particle shape. Rounded boulders and pebbles included in deposits in glaciated areas are one example. Another is the smooth, rounded gravel and sand particles in flowing streams.

Pit or bank-run material is usually screened to proper size, separating sand from gravel and even further separating gravel into different sizes. The larger particles of natural gravel are often crushed to make them more suitable for asphalt paving mixtures. Crushing typically improves the particle shape by making the rounded particles more angular; it also improves the distribution and range of particle sizes. Many times, pit or bank-run material must also be washed to remove dirt (excessive amounts of silt and clay) before using it for asphalt paving.

Quarried Aggregates

Quarried aggregates are also natural aggregates and are produced from solid rock formations. Figure 3.2 shows a typical rock quarry excavation. After processing, the aggregate is commonly referred to as crushed stone.

In the manufacture of crushed stone, solid ledges of quarry bedrock are broken up by blasting and then further reduced in size by rock crushers. The

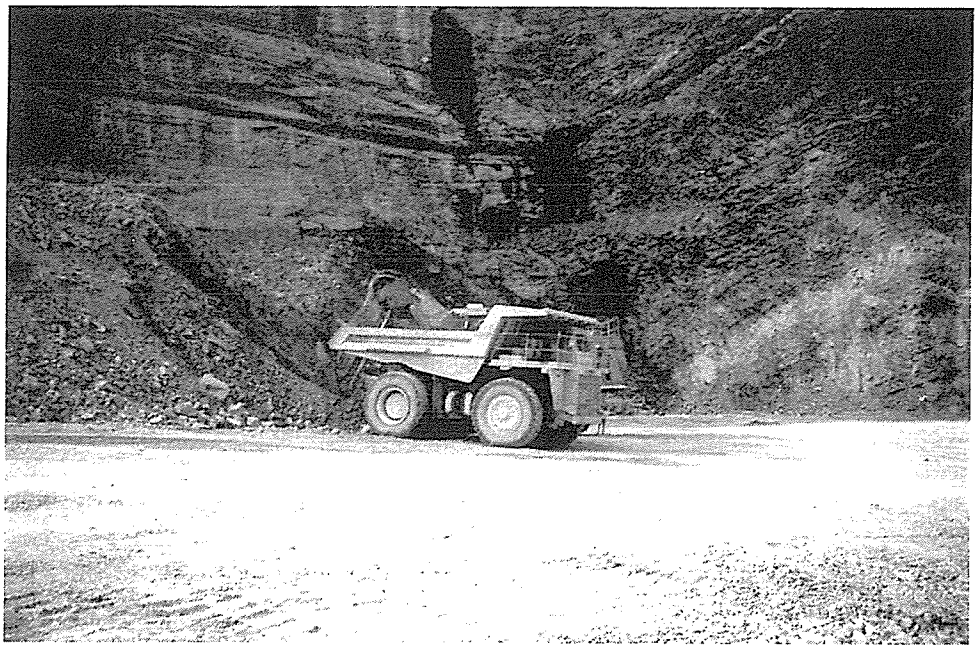


Figure 3.2 Rock Quarry Excavation

resulting crushed stone consists of aggregate particles with fractured faces. Usually, the crushed product is screened to produce the desired aggregate sizes. Sometimes, for cost efficiency, the crushed product is used as it comes from the crusher with little or no screening. This is referred to as "crusher-run" aggregate and, in some instances, can be used satisfactorily in asphalt pavement construction, especially if the graduation of particle sizes is reasonably distributed. Extra care must be exercised, however, to maintain a well-blended product and avoid segregation.

The process of crushing aggregate, whether stone or gravel, generates crushed particles of all sizes. Particles smaller than 0.02 inch (6 millimeters) in diameter are usually separated from the larger crushed material. This fine-aggregate material is known as crushed sand, manufactured sand, crusher fines, and crushed fines. Many times, this material is used in the fine-aggregate component of asphalt paving mixtures. The crushed-sand aggregate from limestone may be further processed to a maximum size of 0.02 inch (0.6 millimeter) in diameter for use as mineral filler in asphalt paving mixtures.

Synthetic and Lightweight Aggregates

Aggregates resulting from the modification of materials, which may involve both physical and chemical changes, are sometimes called synthetic, manufactured, or artificial aggregates. They may be industrial by-products or the result of a specially produced or processed raw material for ultimate use as aggregates. Typically, synthetic aggregates are lightweight and have

excellent resistance to wear. Historically, their preferred use has been in bridge-deck or roof-deck paving and in surface layers of asphalt pavements where a high degree of skid resistance is required.

Slag is a by-product of the iron- and steel-making processes. Slag that is coproduced with iron in blast furnaces is known as Blast Furnace Slag (BFS). Slag that is generated during the production of steel in basic oxygen furnaces or electric arc furnaces is called steel furnace slag or steel slag. For aggregates used in asphalt paving applications, both types of slag are processed by the air-cooled method in which the molten slag is placed into beds and cooled slowly under atmospheric conditions. The hardened, crystalline material is subsequently crushed and screened.

Expanded Shale, Clay, and Slate (ESCS) are often called structural lightweight aggregates. These aggregates are derived from the intense heat treatment of the naturally occurring materials of shale, clay, and slate. The raw materials are subjected to temperatures that exceed 1800–2100°F (980–1150°C) in a rotary kiln, which causes them to expand. These aggregates have a rough texture and are angular. They have a strong affinity for asphalt cement and provide good skid resistance in asphalt paving surfaces. Because of their insulating qualities, underlying subgrades of pavements that contain lightweight aggregates show some evidence of reduced frost depths. Because these aggregates may have densities as low as one-half of conventional aggregates, adjustments are needed in mix-design calculations to accommodate for differences in specific gravity when these aggregates are used in mixes in combination with conventional aggregates.

Many synthetic aggregates are made from reclaimed materials. Some have been used for some time but on a limited basis, while others are in developmental stages. Reclaimed materials used in these synthetic aggregates include recycled plastic and glass, processed sludge and sediment, and by-products from coal combustion.

Recycled glass has been used as an aggregate substitute in HMA paving. The recycled glass is crushed and screened, yielding fine aggregate—typically minus No. 4 sieve (4.75 millimeters) aggregate particles. Coarse glass particles typically contain flat and elongated particles, which need to be avoided. The quantity of flat and elongated particles depends on the crushing equipment and the amount of crushing to which the glass is subjected. Specific gravity of the material is 2.0–2.5, and absorption is obviously insignificant. When recycled glass is used in HMA, glass processing must include the removal of ferrous and nonferrous metal, plastic, and paper. The term *glasphalt* has at times been used to describe pavements containing recycled glass.

There has been evidence in the past of poor bonding with asphalt binder that has resulted in stripping and raveling.

Two other waste materials that have been used as aggregate substitutes in HMA mixtures are ground rubber and roofing shingles. These materials are discussed in Section 3.4, **Aggregate Additives.**